

PAT Preparation Curriculum(EEE)

Aptitude Syllabus

Module 1: Number System

- Factors and Multiples
- LCM and HCF
- Divisibility Rules

Module 2: Algebra

- Linear Equations
- Two-Digit Number Problems
- Algebraic Identities
- Age Problems

Module 3: Percentages

- Percentage Basics
- Reverse Percentage Problems

Module 4: Profit, Loss and Discount

- Cost Price
- Selling Price
- Marked Price
- Discount Calculations

Module 5: Ratio and Proportion

- Ratio Concepts
- Age Ratio Problems

Module 6: Average

- Arithmetic Mean
- Weighted Average

- Marks-Based Problems

Module 7: Time, Speed and Distance

- Relative Speed
- Meeting Problems

Module 8: Time and Work

- Work Efficiency
- Men and Days

Module 9: Probability

- Dice Problems
- Probability Fundamentals

Module 10: Mensuration

- Area of Four Walls
- Cuboid Measurements

Module 11: Clock Problems

- Gain and Loss of Time



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Academic Syllabus

Module 1

Vector Calculus, Transform Techniques

- Advanced calculus operations
- Electromagnetic and signal processing modeling

Microcontroller

- Microcontroller architecture and register organization
- Peripheral interfacing
- Embedded C and assembly implementations (sessional)

- Embedded hardware debugging

Network Analysis

- Core circuit analysis techniques (Mesh and Nodal analysis)
- Transient response modeling
- Network theorems
- Steady-state AC circuit responses

Analog Signal Processing & Laboratory

- Operational amplifier (Op-Amp) configurations
- Active filters
- Signal conditioning circuits
- Performance validation of analog circuits

Generation, Transmission & Distribution

- Layout and architecture of electrical power production systems
- Parameter calculations for overhead transmission lines
- Power distribution mechanisms

Module 2

Probability Theory and Numerical Methods

- Statistical tracking methods
- Numerical approximations for load forecasting and data modeling

Instrumentation and Measurements

- Working principles of measurement systems
- Transducers and digital instrumentation
- Error analysis and hardware calibration setups

Linear Control Systems

- System modeling, block diagram reduction, and signal flow graphs
- Time-domain and frequency-domain stability analysis
- Analytical tools: Routh-Hurwitz, Nyquist plots, Bode plots, and Root Locus

Electrical Machines I & Laboratory

- Operating principles and construction of electrical machines
- Testing and performance characteristics of DC Generators and DC Motors
- Testing and performance characteristics of Single-Phase and Three-Phase Transformers

Electromagnetic Fields

- Static and dynamic electric and magnetic fields
- Maxwell's equations and boundary conditions
- Wave propagation analysis

Module 3

Electrical Machines II

- Advanced operating principles and structural dynamics
- Efficiency equations
- Speed-torque curves of Synchronous and Asynchronous (Induction) machinery

Modern Switchgear and Protection

- Protective relays
- Circuit breaker operational mechanics
- Fault detection networks
- Power grid safety coordination architectures

Power Electronics & Laboratory

- Performance parameters and switching mechanics of power semiconductor devices (SCRs, MOSFETs, IGBTs)
- Power conversion: AC-DC rectifiers and DC-DC buck/boost converters
- Inverters and motor speed control applications

Module 4

Signal Analysis and Processing

- Mathematical signal transformations
- Continuous and discrete signals
- Z-transforms and Fourier analysis
- Digital filter implementations

Power System Analysis and Stability & Laboratory

- Symmetrical and unsymmetrical fault calculations
- Load flow modeling
- Transient and steady-state power system stability
- Numerical optimization software implementations



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